Supplementary Information for:

Worldwide inventory reveals the frequency and variability of pyrocumulonimbus and stratospheric smoke plumes during 2013-2023

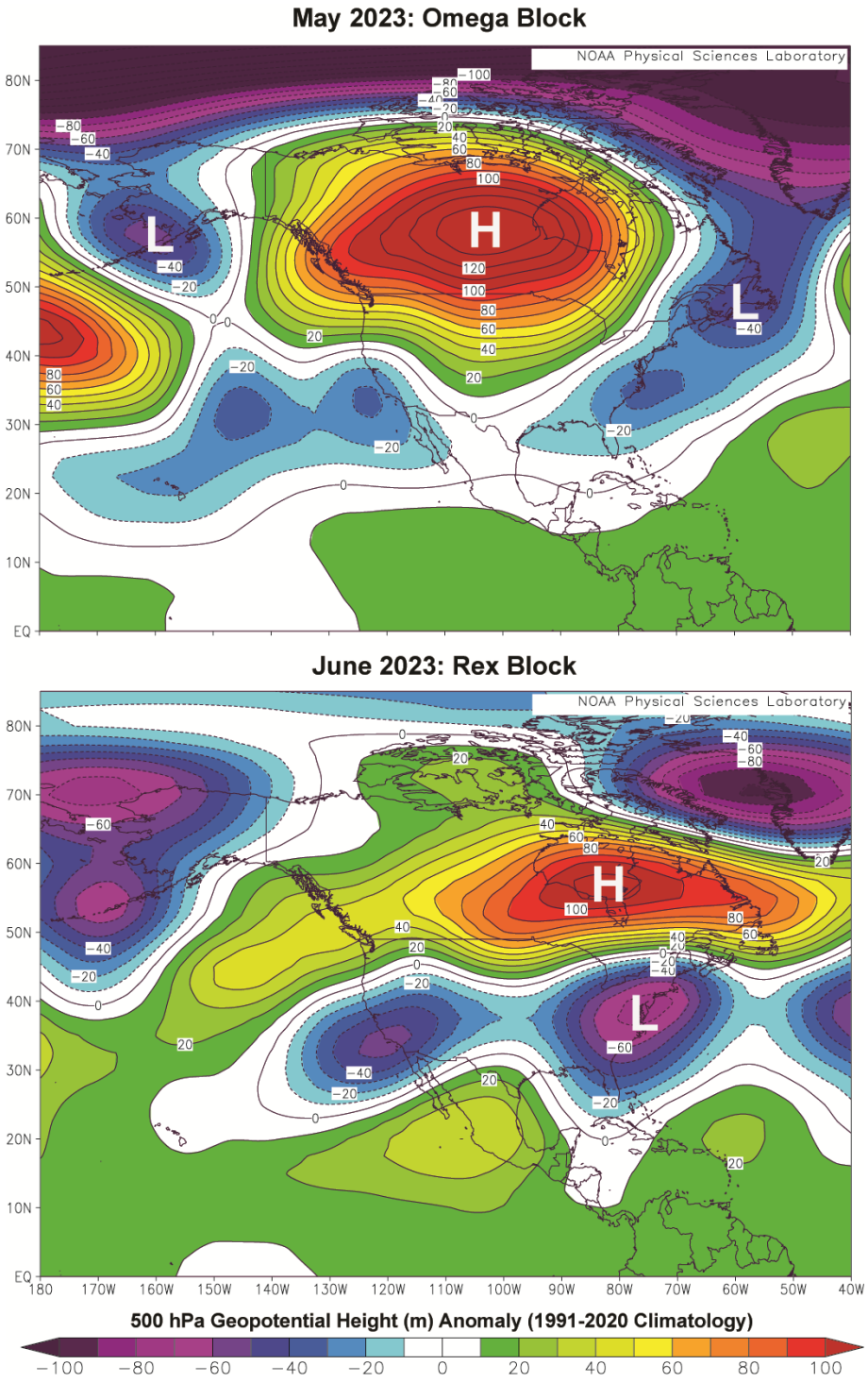
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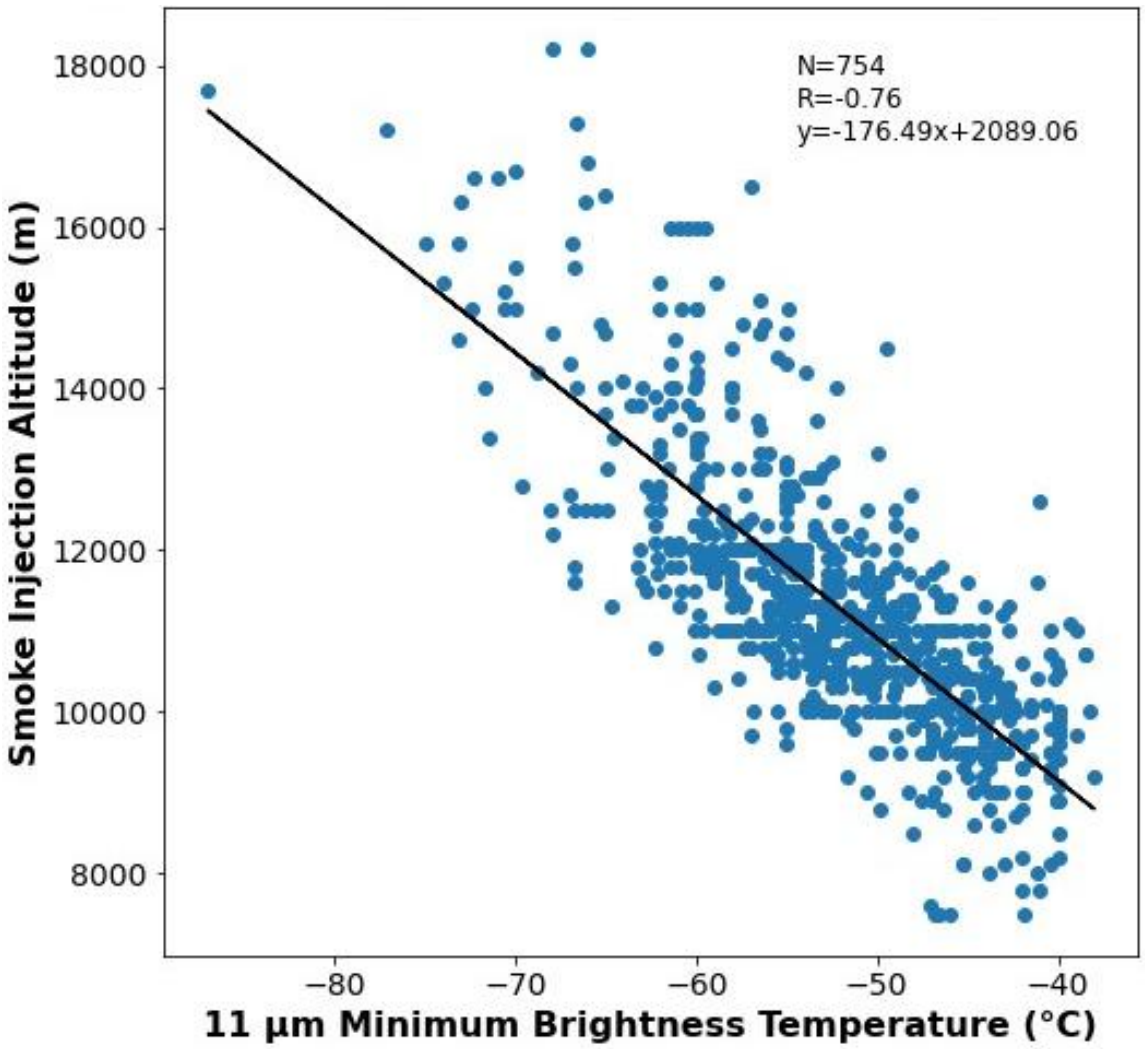
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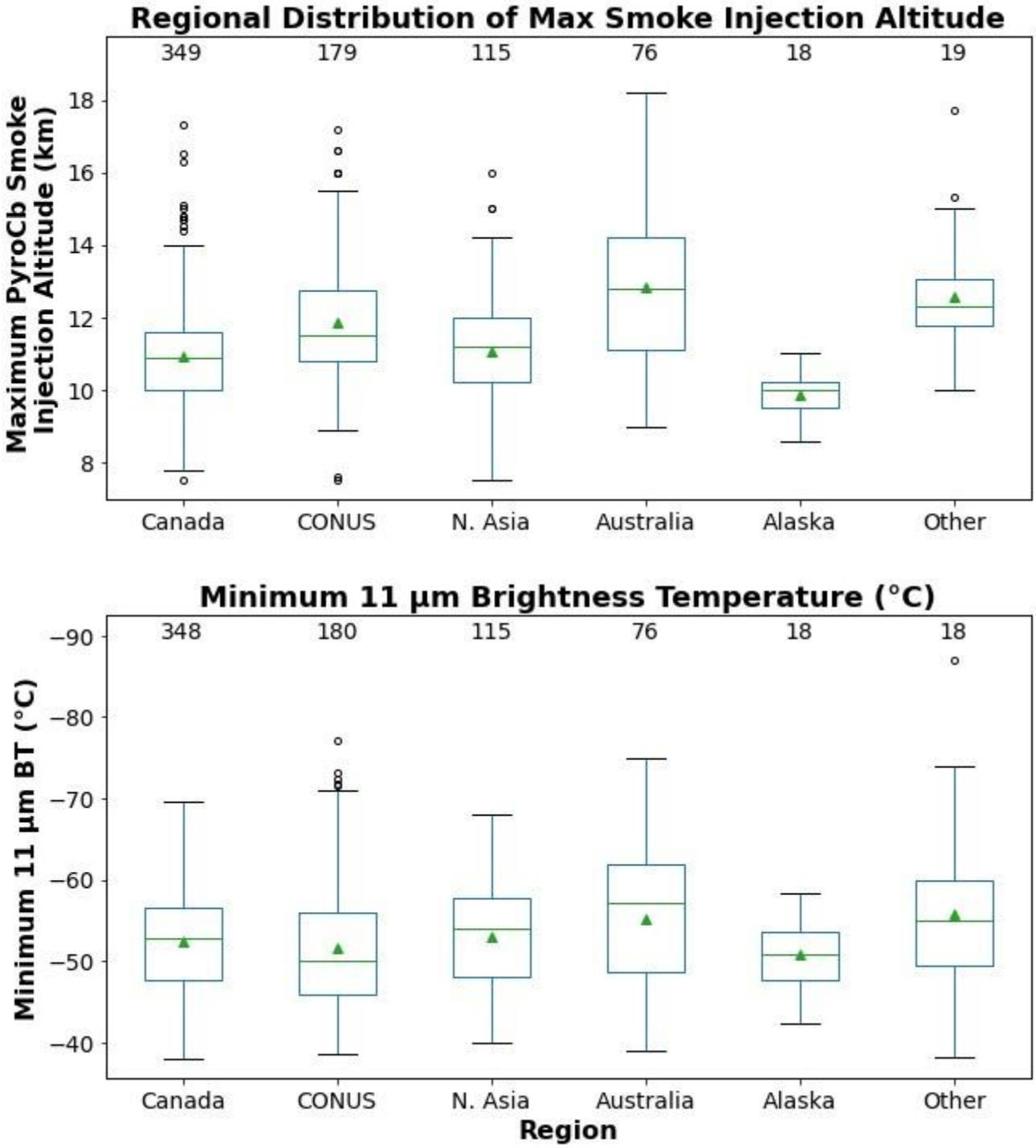
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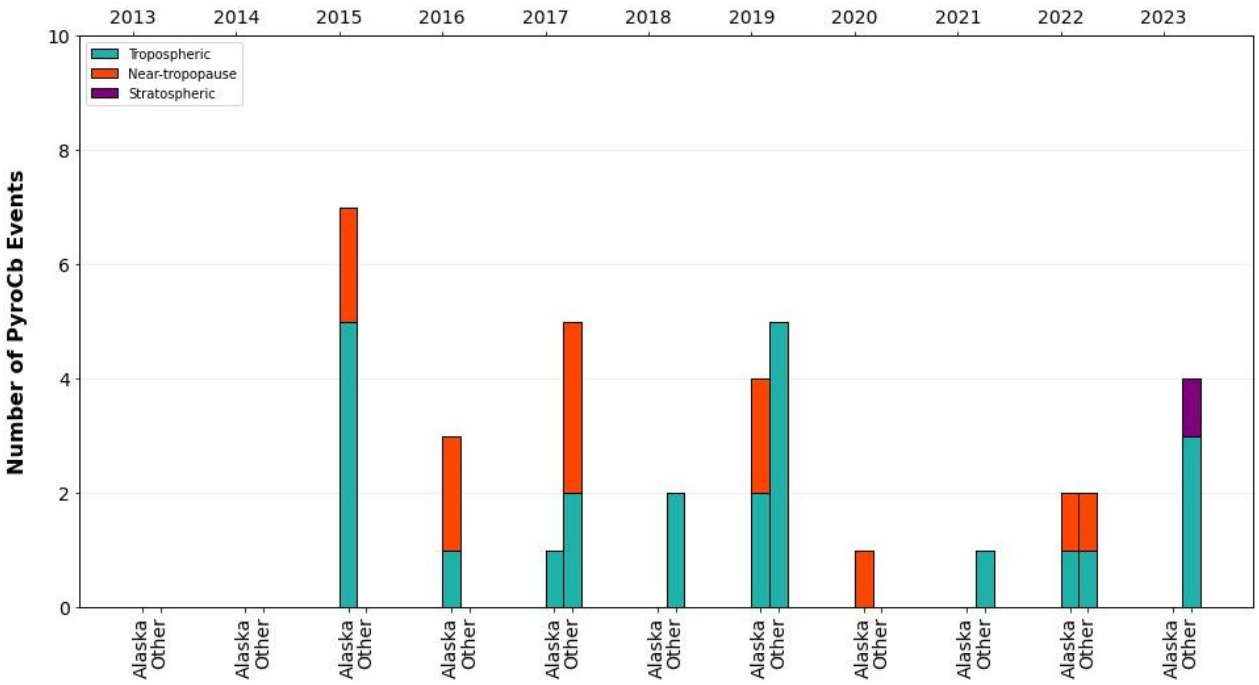
Supplementary Figure 1. Analysis of 500 hPa height anomalies, based on a 1991-2020 climatology from the NCEP/NCAR Reanalysis, for the first two months of the record-breaking fire and pyroCb Canadian fire season in 2023. The large-scale features highlight an Omega Block pattern in May and a Rex Block pattern in June.



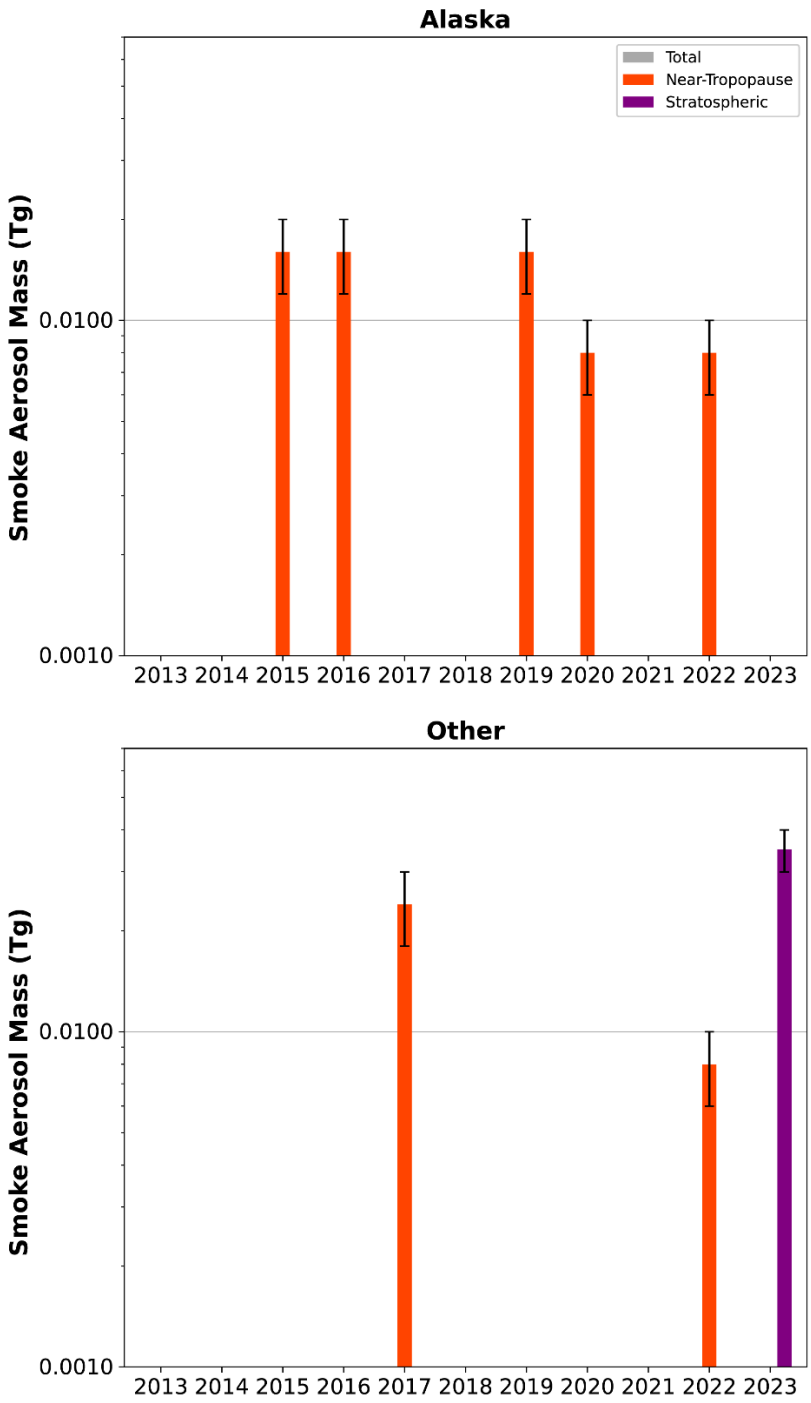
Supplementary Figure 2. Scatterplot comparing minimum BT_{11} to maximum smoke injection altitude for all confirmed pyroCb events with available data during 2013-2023.



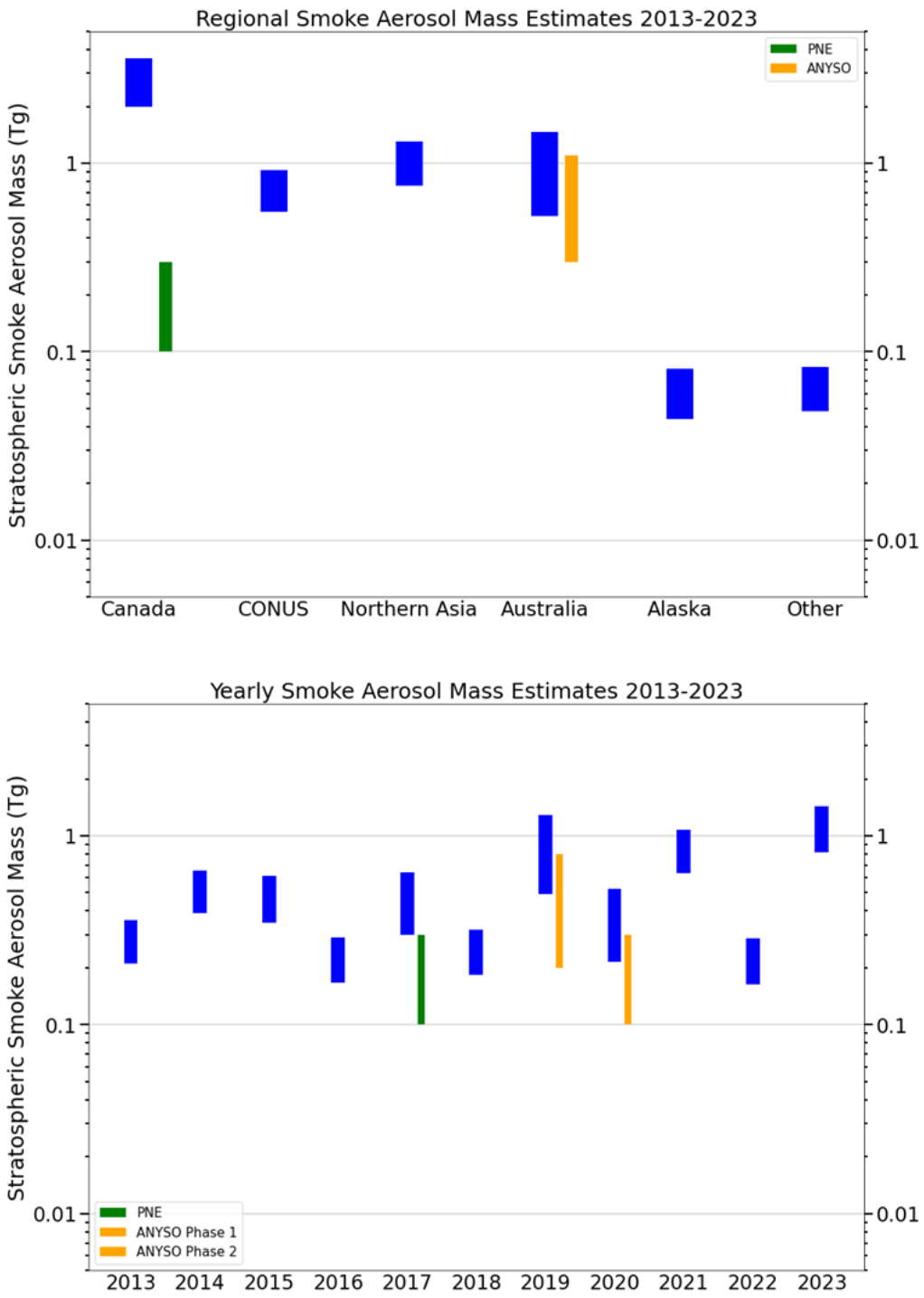
Supplementary Figure 3. Distributions of minimum BT_{11} and maximum smoke injection altitude by region (top) and year (bottom) for all confirmed pyroCb events during 2013-2023. The boxes are bounded by the 25th and 75th percentiles, with the median indicated as a bar bisecting each box. Mean values are displayed as triangles. Data outliers are shown as open circles.



Supplementary Figure 4. Yearly pyroCb maximum smoke injection altitude distributions for Alaska and other regions. Details for each shaded smoke injection altitude category are provided in Fig. 7.



Supplementary Figure 5. Yearly pyroCb smoke aerosol mass estimates for Alaska (top) and other regions (bottom). Injected smoke aerosol mass estimates for each year are divided into stratospheric (purple) and near-tropopause (orange) components, along with the combined total (gray) if applicable. Uncertainty ranges are denoted by black bars.



Supplementary Figure 6. Regional (top) and yearly (bottom) total smoke aerosol mass estimates for all pyroCb plumes in the UTLS worldwide (combined near-tropopause and stratospheric injections), with uncertainty ranges denoted by blue bars. Estimated components of the total smoke aerosol mass attributed to PNE and ANYSO are denoted by green and yellow bars, respectively.

Information on Supplementary Data File 1 (Full PyroCb Inventory)

Supplementary Data File 1 (separate supplementary file) contains the full worldwide pyroCb inventory for 2013-2023. The file includes 793 events in the following categories:

- 761 confirmed pyroCb events used in this study.
- One additional confirmed pyroCb event on 02 September 2023 that was discovered in recent retrospective analyses of pyroCb activity for other studies (see notes column).
- 31 events with a status of “probable”.

Supplementary Tables 1 and 2 provide expanded details for the 15 variables included in the inventory data file for all 793 events.

Supplementary Tables

Supplementary Table 1. Expanded details of variables included in the worldwide pyroCb inventory for 2013-2023.

Variable Name	Units	Expanded Description
Datetime	N/A	Date and time when the pyroCb event started in UTC. Formatted as mm/dd/yyyy hh:mm.
Fire_name	N/A	Obtained from official government web sites. Sometimes based on non-official information such as a nearby location, a forest name, etc. where the pyroCb developed. Multiple names (slash delimited) indicate either uncertainty or merging fires.
Region	N/A	Name of region where pyroCb event occurred: Alaska, Australia, Canada, CONUS (Contiguous United States), Northern Asia (Russian Federation [Siberia], Mongolia and Kazakhstan) and Other (Argentina, Brazil, Bolivia, Democratic Republic of Congo, Paraguay, Portugal, Spain and South Africa).
Status	N/A	C: PyroCb event is confirmed. P: PyroCb event is probable. "C" indicates the evaluation is completed and all criteria for a pyroCb are met. "P" indicates the evaluation is completed and this is likely a PyroCb, bot all criteria are met to confirm due to one or more factors hindering detection (e.g., nighttime onset, meteorological clouds, lack of satellite data).
Lat	Decimal degrees	Approximate latitude of fire linked to pyroCb development. Positive (negative) for Northern (Southern) Hemisphere.
Lon	Decimal degrees	Approximate longitude of fire linked to pyroCb development. Positive (negative) for Eastern (Western) Hemisphere.
Pcb_min_bt	Degrees Celsius	Minimum 11 µm brightness temperature (BT_{11}) detected in the pyroCb anvil and obtained from various satellites.
Trop_alt	Meters	Tropopause altitude derived from MERRA-2.
Max_inject_alt	Meters	Maximum pyroCb smoke injection altitude assigned manually, primarily using a combination of radar echo-top data, BT_{11} from multiple satellite sensors, and vertical profiles from ground and spaceborne lidar.
Trop_relative_alt	Meters	Maximum pyroCb smoke injection altitude relative to tropopause altitude. Negative when maximum pyroCb injection altitude is below the tropopause.
Min_smoke_mass	Teragrams	Smoke aerosol mass estimate lower bound for pyroCb plumes injected within one kilometer of the tropopause or higher.
Max_smoke_mass	Teragrams	Smoke aerosol mass estimate upper bound for pyroCb plumes injected within one kilometer of the tropopause or higher.
Plume_code	N/A	Identifies if smoke aerosol mass estimate ranges are obtained from analyses of well-observed pyroCb plumes or estimated based on statistics of other well-observed pyroCb plumes in the same injection altitude category. Full details are available in Supplementary Tables 2 and 3.
Pulse_code	N/A	Binary. "0" for a single-pulse pyroCb, "1" for a multi-pulse pyroCb.
Notes	N/A	Supplemental information, including references to publications, information on the magnitude of the pyroCb event, etc.

Supplementary Table 2. Plume code information.

Plume Code	Aerosol Mass Estimate	Description
indv_well_obs*	Calculated directly from remotely-sensed plume observations.	Individual pyroCb event with well-observed plume information in the near-tropopause or stratospheric injection categories.
outb_well_obs_NAME*	Calculated directly from remotely-sensed plume observations.	PyroCb outbreak with well-observed plume information in the near-tropopause or stratospheric injection categories. Upper and lower bounds provided for pyroCb outbreaks represent the smoke aerosol mass injected by all pyroCb activity in each outbreak. The values should only be applied once for each outbreak, regardless of how many pyroCb events are included. Outbreak name is included in the 'plume_code' string.
indv_est	0.006-0.01 Tg	Individual pyroCb event with injected smoke aerosol mass estimated based on statistics of other well-observed pyroCb plumes primarily in the near-tropopause injection altitude category.
indv_est	0.03-0.04 Tg	Individual pyroCb event with injected smoke aerosol mass estimated based on statistics of other well-observed pyroCb plumes in the stratospheric injection altitude category.

* Plume aerosol mass estimates are available in Fig. 10 and Supplementary Table 3.

Supplementary Table 3. Full details of the well-observed pyroCb smoke plumes in Fig. 10.

Plume Number	Plume Name in Fig. 10	Date of PyroCb Activity	Number of Contributing PyroCb Events	Minimum BT_{11} (°C)	Maximum Smoke Injection Altitude (km)	Injection Altitude Relative to Tropopause (km)	Minimum Smoke Aerosol Mass Estimate (Tg)	Maximum Smoke Aerosol Mass Estimate (Tg)	Multi-Pulse Event
i	2019 ANYSO Phase 1, Australia*	29-31 Dec. 2019	14	-73.2	16.30	2.80	0.2	0.8	yes
ii	2020 ANYSO Phase 2, Australia*	04 Jan. 2020	4	-70	16.70	0.96	0.1	0.3	yes
iii	2017 PNE, Canada*	12 Aug. 2017	4	-65	13.70	1.30	0.1	0.3	yes
1	2021 Sparks Lake, BC, Canada*	30 Jun. 2021	2	-66.6	17.30	3.58	0.025	0.05	yes
2	2017 Outbreak, Northern Asia*	22 Jun. 2017	3	-58.5	12.50	0.08	0.017	0.038	no
3	2017 Diamond Creek, CONUS	29 Aug. 2017	1	-56.2	14.80	3.27	0.028	0.034	yes
4	2020 Creek Fire, CONUS	05 Sep. 2020	1	-71	16.60	0.69	0.0065	0.03	yes
5	2017 Canada 1	04 Sep. 2017	1	-54.7	12.80	0.11	0.011	0.022	no
6	2017 Northern Asia	15 Aug. 2017	1	-68.1	12.50	0.02	0.012	0.021	no
7	2020 Northern Asia 1	09 Jun. 2020	1	-53.6	10.20	0.30	0.0077	0.012	no
8	2013 Sliver Fire, CONUS	28 Jun. 2013	1	-70.6	15.20	0.43	0.0045	0.0078	yes
9	2017 Canada 2	23 Jun. 2017	1	-55	9.80	-1.34	0.0034	0.0069	yes
10	2020 Northern Asia 2	22 Jun. 2020	1	-64.7	11.30	-0.92	0.002	0.0068	no
11	2017 Canada 3	19 Jun. 2017	1	-56.9	10.00	-1.02	0.0021	0.0041	no
12	2013 Canada	03 Jul. 2013	1	-59.6	13.00	0.54	0.00061	0.00072	no

* Smoke aerosol mass estimates for pyroCb outbreak plumes are aggregated over all contributing pyroCb events.